

Beyond Borders: How Voters Process Comparative Information

Adam Roberts

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- Accountability: voters use elections to reward or punish incumbents based on performance
- For accountability to work, voters must be able to:
 - ▶ Observe what the government has done
 - ▶ Attribute outcomes to incumbent actions
 - ▶ Act on that assessment at the ballot box

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- Attribution is difficult when government performance is noisy

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- Many potential reasons for mixed findings
- Relatively less attention on how the information is presented

Does the way the information is presented matter?

- Studies typically vary *whether* information is provided, not *how* it is framed
- Real-world campaigns routinely use comparisons to other subnational governments (Grossman and Slough 2022)
- How do comparisons change interpretation of information?



FINANCIAL MANAGEMENT

In 2013, the accounts of the mayor of Abreu e Lima were APPROVED, as happened in 88% of the municipalities of Pernambuco.

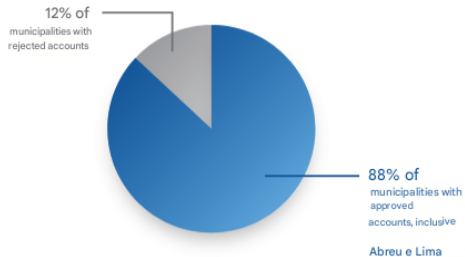


Figure 1: Informational leaflet from Boas, Hidalgo, and Melo (2019)

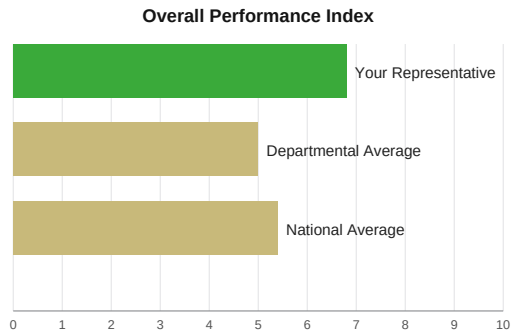
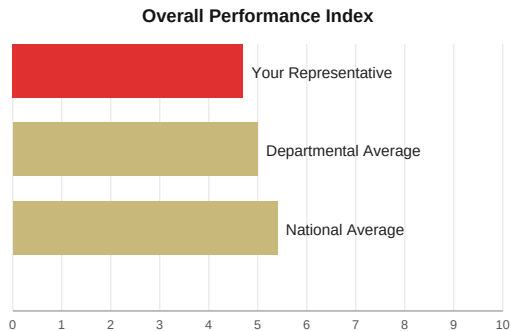


Figure 2: Informational leaflet from Adida et al. (2020)



Anyá pay ti karkarida?

Angel Miguel Hernando (LP)

- Providing free elementary and high school education
- Implementation of Health programs to provide free medicines and Philhealth; improvement of health stations in the villages
- Suspension of tax payment by owners of small business premises (sari-sari stores)

Melanie Valdez (IND.)

- Establishing community based projects
- Introduction of organic farming to integrate agriculture and the environment and to increase yields; making those so-called solid wastes into fertilizer
- Continue to provide free High School and College education through Binhi Foundation scholarships

Figure 3: Information from Cruz, Keefer, and Labonne (2021)

How might voters use comparative information?

1. **Filtering noise:** comparisons clarify incumbent signals (Besley and Case 1995)
 - ▶ Relative to similar politicians and/or those with similar extenuating circumstances
2. **Constructing counterfactuals:** incumbent to opposition comparisons
3. **Unhelpful complexity:** comparisons confuse more than help (Arias et al. 2024)

Survey experiment in Mexico

Two components in one survey:

1. **Descriptive:** respondents select municipalities they consider relevant comparisons
2. **Experiment:** experimentally vary comparisons

Context:

Provide real crime information, electoral parties/coalitions

Survey deployed in subset of states holding municipal elections in 2027

Why crime information?

- Municipal forces comprise majority of law enforcement (Marshall 2024)
- Municipal presidents control police funding, hiring, and structure
- Crime is a top voter concern for Mexicans
- Imprecise priors about local crime → information should update beliefs

Data: robbery rate per 100,000 people in 2025, from Public Security Secretary (SESNSP)

Two-wave survey experiment

Wave 1

Comparison selection

Respondents select comparison municipalities

Pre-treatment beliefs

Crime handling
Coalition ratings
Robbery estimate
Vote Intention

Wave 2

Treatment

Given crime information

Post-treatment

Crime handling
Coalition ratings
Robbery Estimate
Vote intention

Descriptive Results from Wave 1

How do voters choose comparison municipalities?

- 15 candidate municipalities shown before treatment
- Select any they would use as a comparison to their own
- Candidates drawn from three pools: nearby, nationally largest, random
- Bayesian multilevel logistic regression to analyze selection patterns

Municipal governments are often evaluated by comparing them to others. Which of the following municipalities would you compare Monterrey, Nuevo León to? (Choose at least one.)

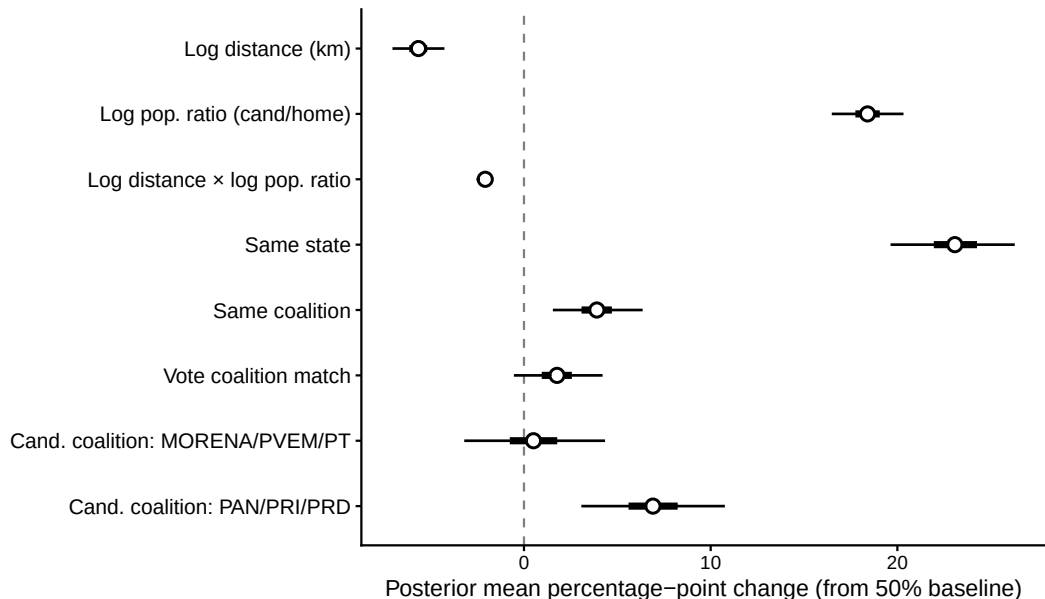
- ☐ Cuautla, Morelos
- ☐ Toluca, Mexico
- ☐ San Nicolás de los Garza, Nuevo León
- ☐ San Pedro Garza García, Nuevo León
- ☐ Juárez, Chiapas
- ☒ Ecatepec de Morelos, Mexico
- ☐ Susupuat, Michoacán
- ☒ Santiago, Nuevo León
- ☐ Puebla, Puebla
- ☐ Reynosa, Tamaulipas
- ☐ Guadalupe, Nuevo León
- ☐ Del Nayar, Nayarit
- ☐ El Carmen, Nuevo León
- ☐ Hermosillo, Sonora
- ☐ Tepechitlán, Zacatecas

Bayesian multilevel logistic regression

$$\Pr(Y_{ri} = 1) = \text{logit}^{-1}(X_{ri}\beta + \alpha_r)$$

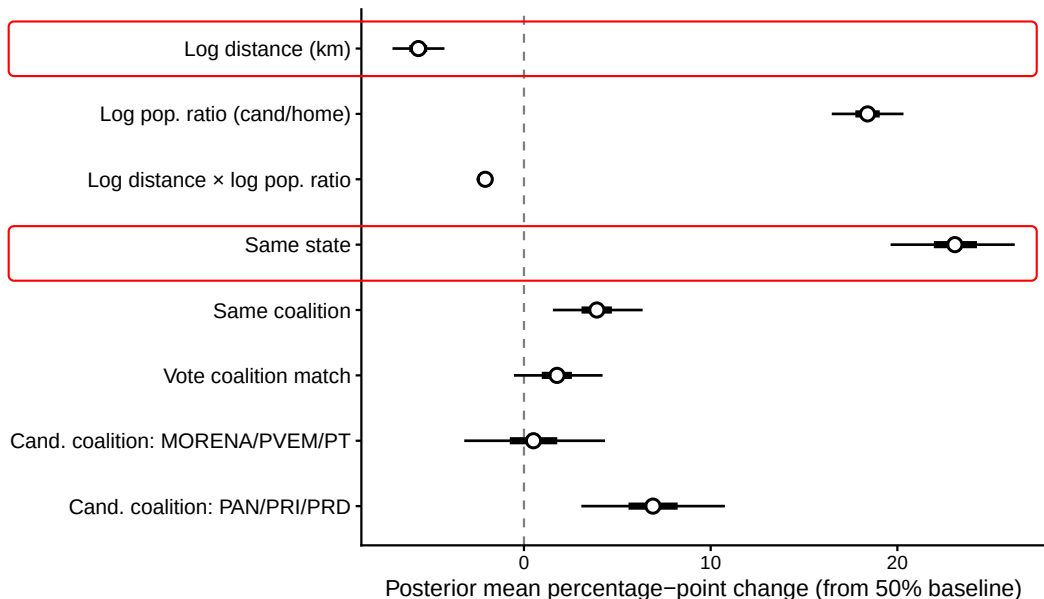
- $Y_{ri} = 1$ if respondent r selects municipality i as a comparison
- X_{ri} : vector of respondent-municipality covariates (e.g., distance, population, same state, same coalition)
- $\alpha_r \sim \mathcal{N}(0, \sigma_\alpha^2)$: respondent-level random intercept

Predictors of comparison municipality selection



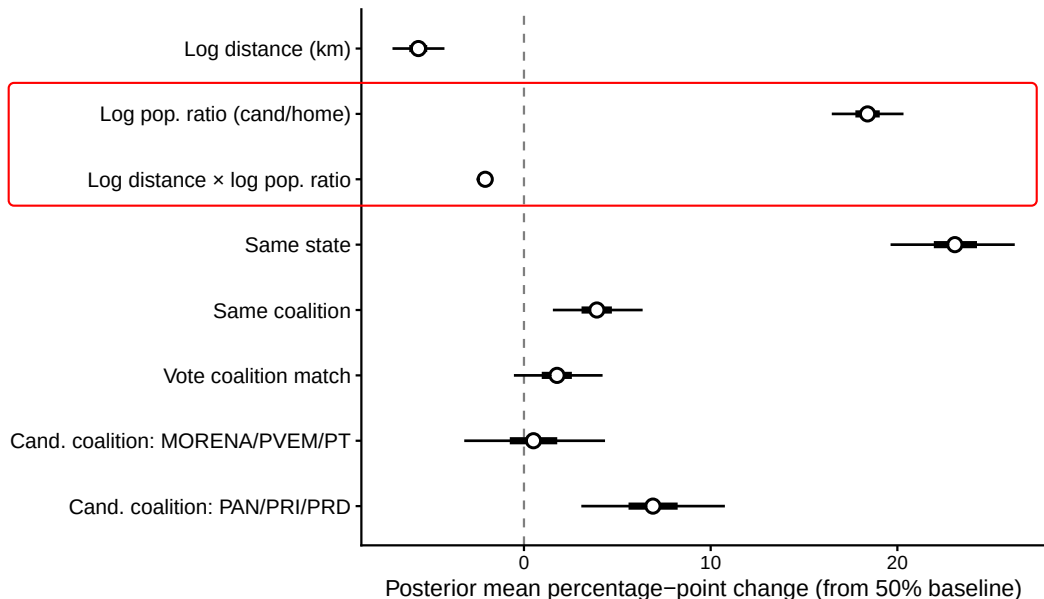
N = 1800 respondents. Thick lines: 50% CI. Thin lines: 95% CI.

Predictors of comparison municipality selection



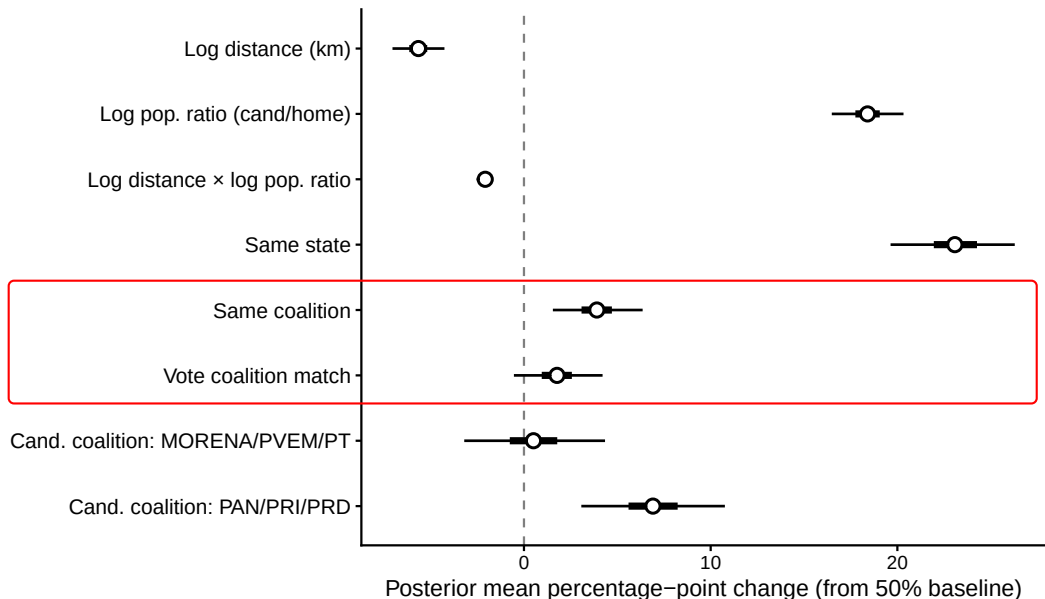
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Predictors of comparison municipality selection

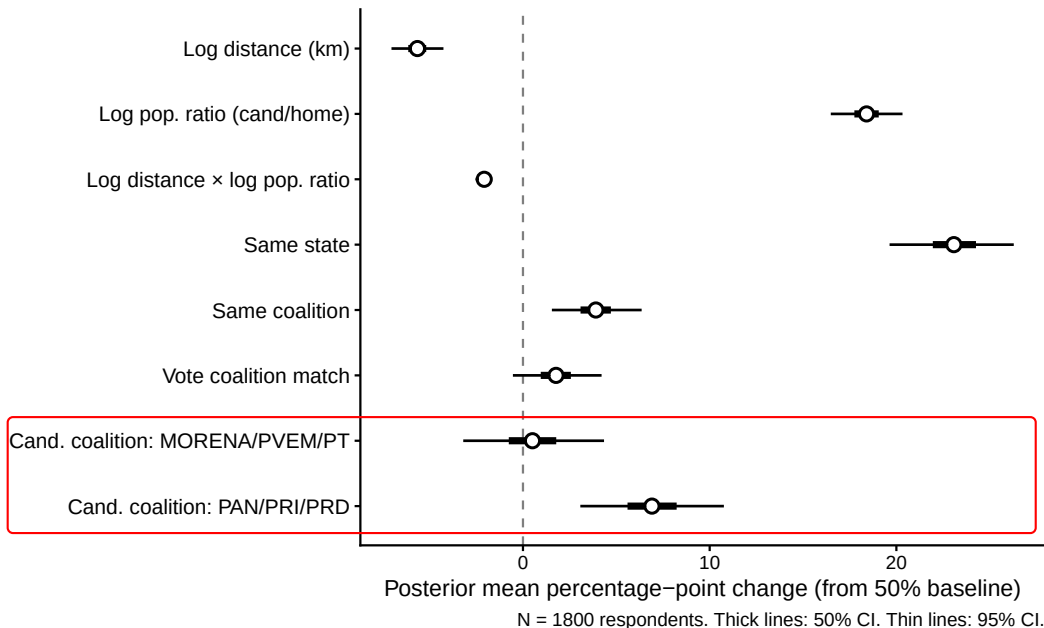


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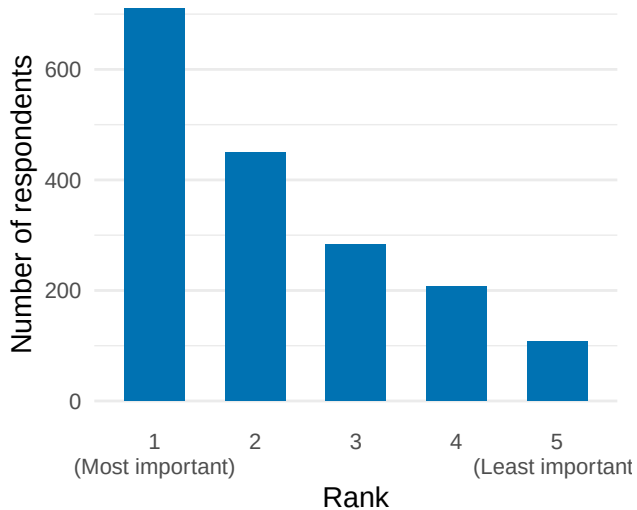
Issue importance question

How important are the following issues to you? Rank them from most to least important.

- Security / Crime
- Economy / Inflation
- Unemployment / Low wages
- Corruption
- Education and health services

(Items presented in randomized order.)

Distribution of Crime Importance



N = 1760

Crime handling survey question

“The following questions will ask you to evaluate how certain municipalities and parties ‘handle’ non-violent crime, such as robberies. ‘Handling crime’ here refers to government efforts to prevent crime, enforce the law, and ensure public safety.”

“How well do you think the government of *[municipality]* — and each major coalition — handles crime?”

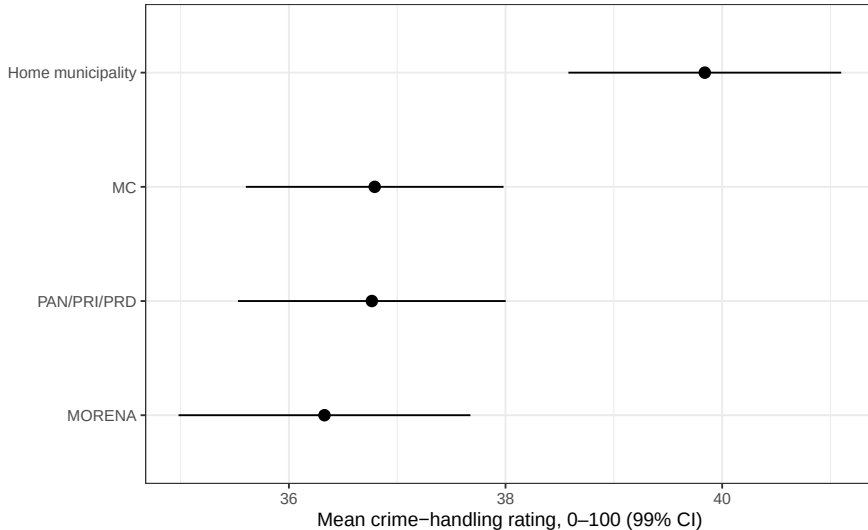
Separate 0–100 slider for each

0 = Handles crime extremely poorly

100 = Handles crime extremely well

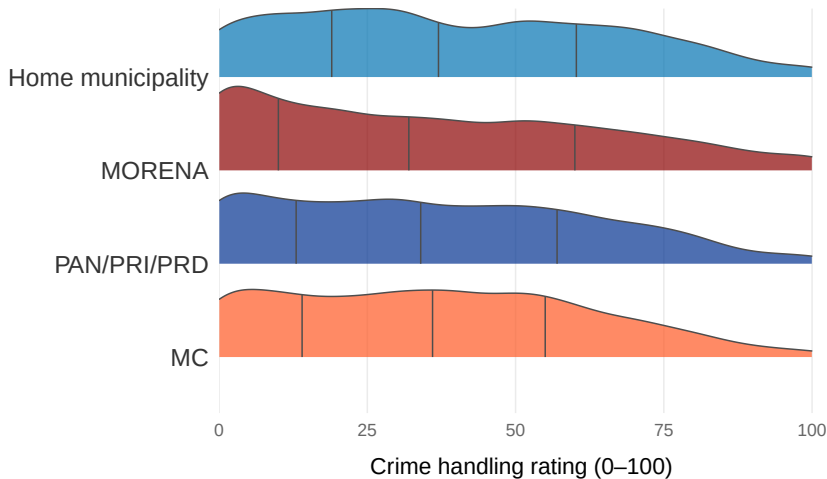
Three major coalitions: MORENA/PVEM/PT, PAN/PRI/PRD, and MC Municipal coalitions

Priors about crime handling



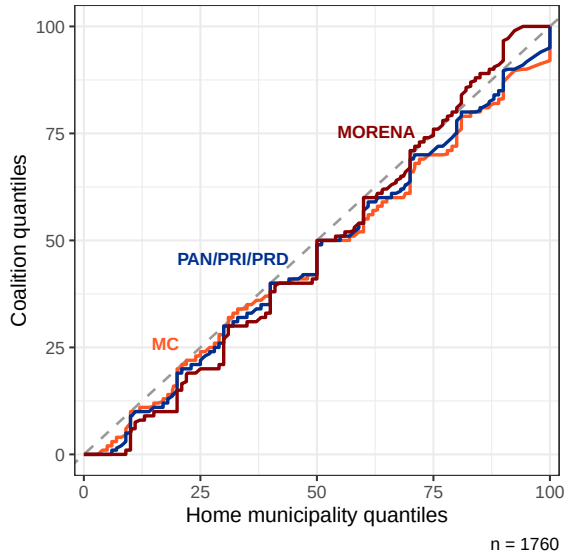
n = 1760

Priors about crime handling

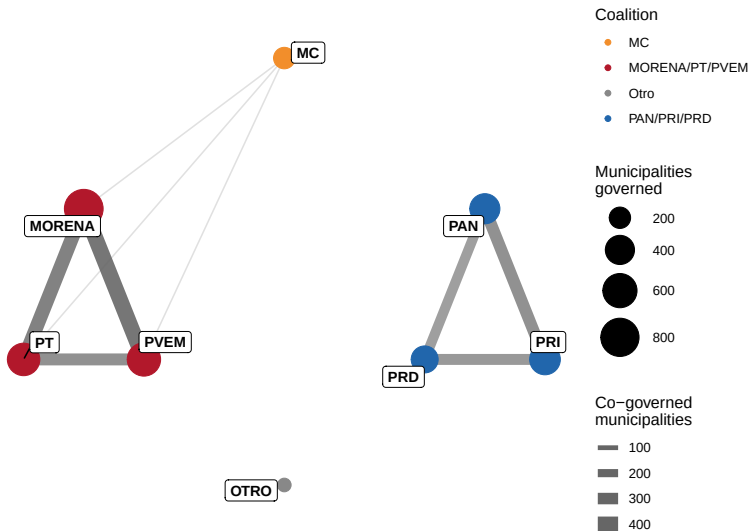


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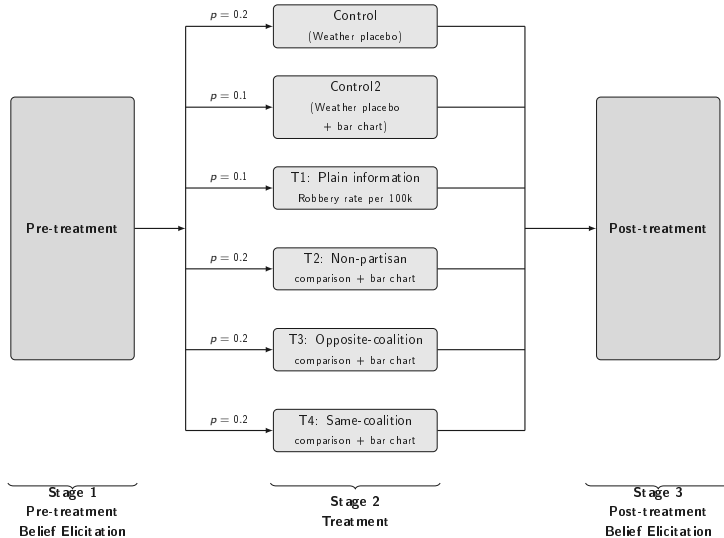
Priors about crime handling



Appendix



Survey design



Example: Toluca's four closest opposition matches

Toluca, Edo. Mex. (MORENA/PVEM/PT) — Pop. 910,608 — Area 424.78 km²

Rank	Match municipality	Population	Area (km ²)	Mahal. dist.
1	Cuernavaca (Morelos)	378,476	198.55	0.82
2	Zinacantepec (Edo. Méx.)	203,872	308.79	1.04
3	Lerma (Edo. Méx.)	170,327	229.44	1.18
4	Cuautla (Morelos)	187,118	121.13	1.41

All four matches governed by PAN/PRI/PRD.

Matching uses Mahalanobis distance on $\log(\text{population})$, $\log(\text{area})$, and $\log(\text{distance})$.

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Treatment text: Control (Weather placebo)

“Geography, altitude, and regional climate systems are the main determinants of precipitation patterns, factors that we largely cannot control. **In [respondent’s municipality] there were X mm of annual precipitation in 2025.**”

Source: WorldClim v2.1 (shown as caption).

Note: Control receives weather placebo text with home municipality precipitation. No crime data or bar chart.

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Treatment text: Control2 (Weather placebo + bar chart)

[Weather placebo text] +

“The following graph shows the average annual precipitation in [respondent’s municipality] and a sample of similar municipalities.”

Source: WorldClim v2.1 (shown as caption below figure).

Note: Control2 mirrors T2 in structure but uses precipitation rather than crime data.

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